



COMPLETE-STACK BENCHMARK - RECONCILED EDITION

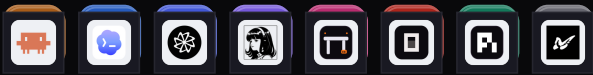
The Stack Is the Product.

What 5,233 hidden-oracle runs say about model x harness pairing, repeatability, cost, and the difference between easy-looking work and genuinely hard engineering.

MODELS



HARNESSES



5,233

VERIFIED RUNS

1,247 model x harness
x task cells

51% vs
12%

HARNESS VS MODEL
SHARE

Observed stack-rate
variance

25.0%

MIXED REPEATED
CELLS

Same setup, both pass
and fail

126-161x

COST SPREAD

Two defensible
weighting rules

25 / 33

GLOBALLY COMMON
TASKS

Default for cross-
model value

HOW TO READ THIS EDITION

Every result is computed from the shipped cell-level dataset. When two defensible definitions answer different questions, both are shown instead of selecting the friendliest number.



01 EXECUTIVE SUMMARY

The one-minute version

The benchmark's most durable lesson is not a winner. It is a change in the unit of analysis: a coding agent is a stack whose model, harness, task mix, and execution behavior must be evaluated together.

01 Buy the stack, not the model.

ROBUST

Harness choice accounts for 51.2% of observed between-stack variance; model choice accounts for 12.4%. On the comparable 25-task set, the ordering remains 54.1% versus 15.5%.

02 Harness rankings are local.

QUALIFIED

Cross-model agreement is positive but only weak-to-moderate: mean Kendall tau-b ranges from 0.288 to 0.423. A recommendation must name the model.

03 Repeatability matters - but $r = -0.95$ is not a new law.

QUALIFIED

308 of 1,230 repeated cells have mixed outcomes. The raw flake/pass correlation is mostly induced by the metric itself; repeatability remains a valuable smoke detector.

04 Price is weakly connected to correctness.

ROBUST

The proxy cost-per-pass spread is 126x under the specified trial-weighted median and 161x under task-equal weighting. On common tasks it is still 115.5x.

05 A green suite can hide hard failure.

QUALIFIED

Main-to-hard rank agreement is tau-b = 0.139 under strict verification, but 0.350 under relaxed timing. Five hard tasks are informative, not a universal population.

06 Use the judge for craft, not truth.

QUALIFIED

At stack level judge/pass correlation falls from 0.922 on the main suite to 0.321 on the hard band. At cell level it barely moves: 0.570 to 0.543.

THE ONE-SENTENCE TAKEAWAY

Stop asking which model is best. Ask which model x harness pairing completes your work repeatedly, under executable verification, at an acceptable cost.



02 THE VERDICT

Three answers, depending on what you're paying for.

SHIPS THE MOST



Codex × Muse Spark 1.2

98.59%

solution rate · 95% CI 95.0%-99.6%

STRICT RATE

98.59%

MEDIAN COST / TASK

\$1.33

RUNS · TASKS

142 · 32

BEST VALUE



Droid × Muse Spark 1.2

\$0.33

median cost / task, at 97.08% solved · standard-tier price

BEHIND THE LEADER BY

1.51 PTS

STRICT RATE

96.35%

RUNS · TASKS

137 · 33

BEST CODE QUALITY



Codex × DeepSeek V4

4.5/5

blind judge median over 97 judged trials

SOLUTION RATE

94.00%

MEDIAN COST / TASK

\$0.0252

RUNS · TASKS

150 · 32

The Muse Spark contributor tier is \$0.00 under data-sharing. Cards show standard-tier pricing - what you'd actually pay.

The gap between the leader and the median stack is 9.2 points of tasks solved. Picking the stack is worth more than picking the model.



03 DESIGN AND COVERAGE

What was actually measured

BurnBench crosses five models with eight agent harnesses over 33 measured software-engineering tasks. Most cells ran three to five times. Pass/fail is decided after the agent exits by hidden test oracles that the agent never sees.

COVERAGE IS NOT COMPLETE

Only **25 of 33 tasks** appear in all 40 stacks. Cross-model cost and capability claims therefore default to the global common-25 set; broader as-run views remain useful but composition-sensitive.

MODEL	TASKS / HARNESS	BALANCED	RUNS	HARD BAND
DeepSeek V4	32-32	32	1,186	Measured hard band
GLM 5.2	31-33	30	1,113	Measured hard band
GPT 5.6	32-32	32	1,155	Measured hard band
Muse Spark 1.2	32-33	31	1,134	Measured hard band
Qwen3.7 Flash	26-28	26	645	Hard band not run

Measured tasks per stack

DeepSeek V4	32	32	32	32	32	32	32	32
GLM 5.2	33	32	32	32	32	33	31	31
GPT 5.6	32	32	32	32	32	32	32	32
Muse Spark 1.2	32	32	33	33	33	33	33	33
Qwen3.7 Flash	28	27	27	27	27	28	26	27
	CC	CX	DR	HE	OM	OC	PI	PA

Numbers are measured tasks in each model x harness stack. The hard band is absent for Qwen.

Strict rate

Hidden oracle passes cleanly. Headline outcome.

Solution rate

Strict passes plus verifier-accepted relaxed timing.

Mixed cell

A repeated cell where 0 < strict passes < trials.

Proxy cost / pass

Median task cost divided by pooled strict rate; not observed successful-run cost.

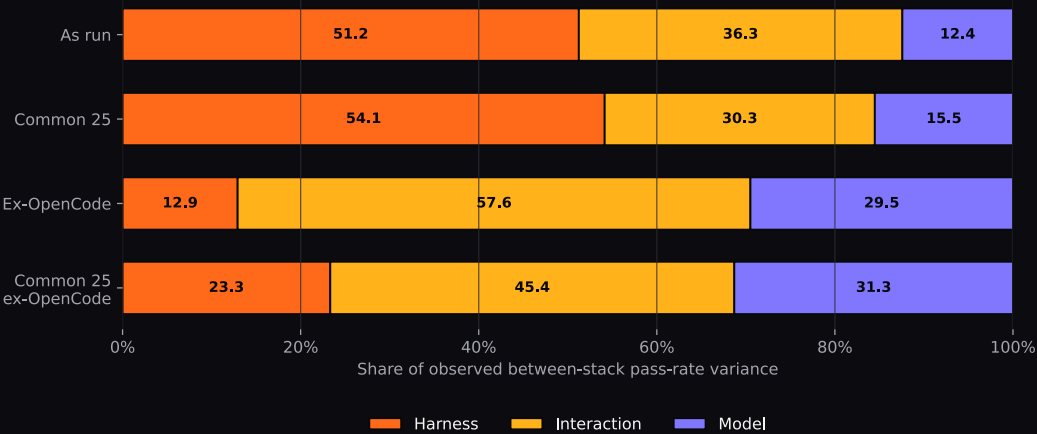


04

WHERE PERFORMANCE LIVES

The stack is the unit that moves

Across the 40 observed stack means, the harness main effect is four times the model main effect. Removing OpenCode changes which term dominates, but not the purchasing lesson: the specific pairing remains too large to infer from model reputation alone.



Shares describe this 5 x 8 matrix. They are not causal population estimates; the interaction term also contains sampling noise.

51.2% harness

As-run share of observed stack-rate variance.

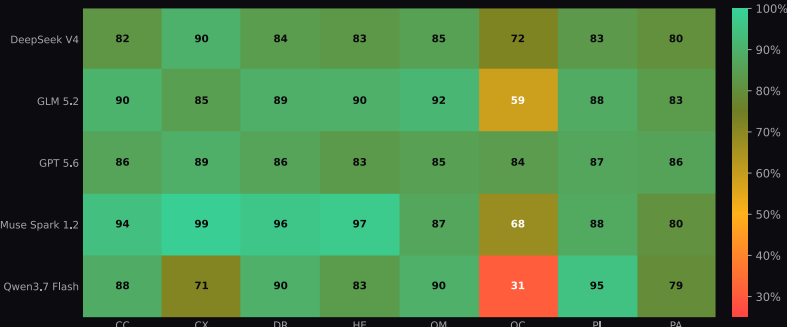
36.3% pairing

Model x harness interaction; 45.4% on common tasks without OpenCode.

12.4% model

15.5% on the 25 tasks shared by every stack.

Balanced within each model: strict pass rate (%)



Balanced within each model. The same model can move by tens of points when the harness changes.

DECISION

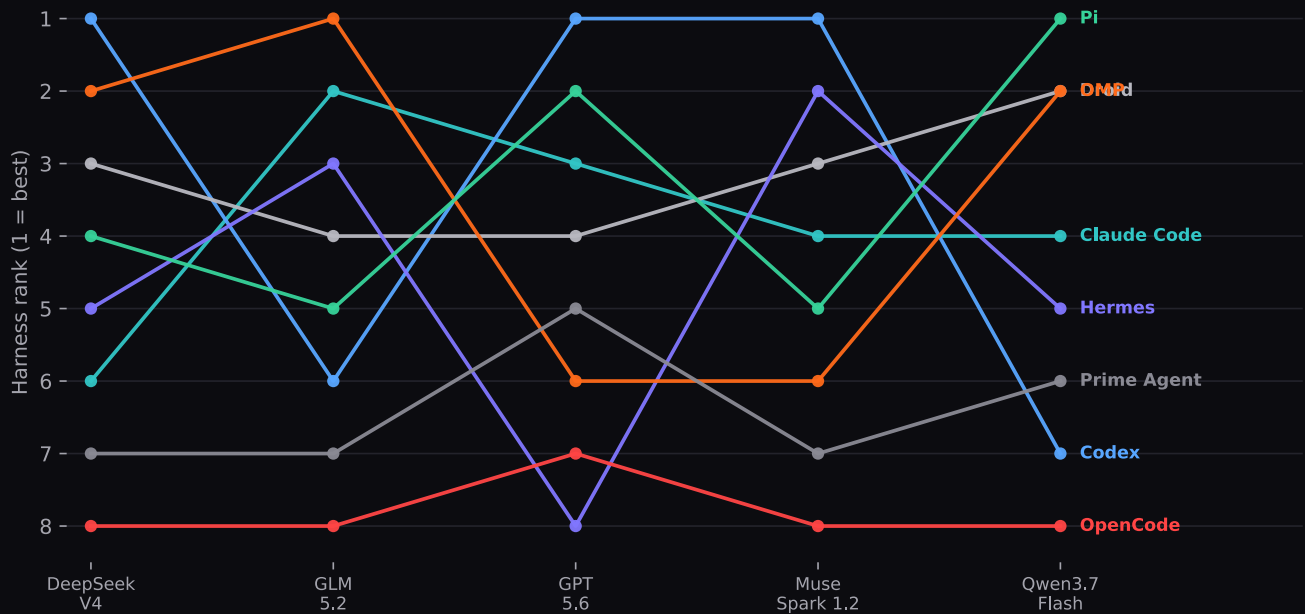
Approve budgets for **model + harness + task family**, not for a model name by itself. The full matrix is the procurement object.



05 RANK TRANSFER

There is no universal best harness

The same eight harnesses, ranked by five models



Ranks use each model's own balanced task set. Lines crossing is the result: harness quality is conditional on the model.

0.288

AS-RUN TAU-B

Broadest, composition-sensitive

0.329

PAIR-BALANCED TAU-B

Preferred transfer estimate

0.423

COMMON-25 TAU-B

Narrower, fully comparable

READ THE EXTREMES

Codex ranks first for DeepSeek, GPT 5.6, and Muse Spark - but seventh for Qwen. Hermes ranks second for Muse and eighth for GPT. Pi ranks first for Qwen.

OPENCODE, HONESTLY

It ranks last for four models and seventh for GPT as run. Its deficit is concentrated in no-tool-call runs, but exact executed-only rates are **bounded rather than identified** because pass/noop overlap is absent from the cell export.



06 REPEATABILITY

Measure it. Do not mythologize it.

308

MIXED CELLS
of 1,230 repeated cells

-0.950

RAW FLAKE X PASS R
Descriptively near-perfect

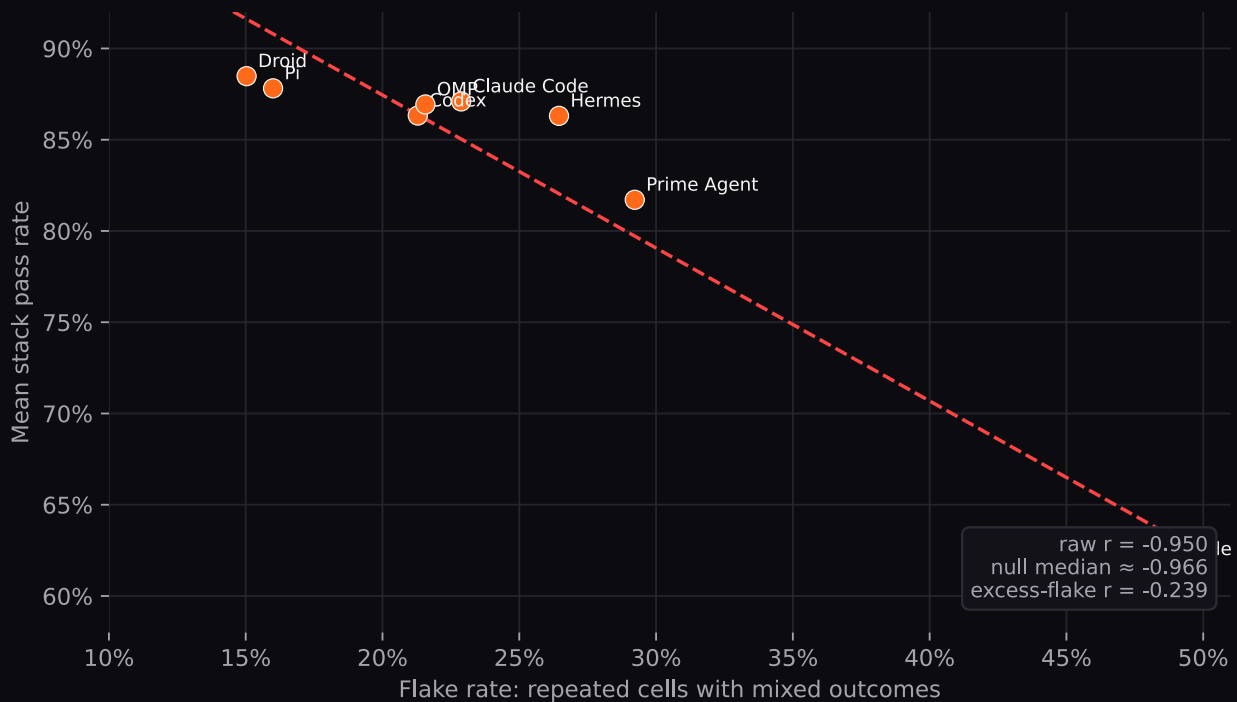
-0.966

NULL MEDIAN
No separate reliability property

-0.239

EXCESS-FLAKE R
After mechanical expectation

Repeatability is operationally useful; the raw correlation is mostly structural



A setup at 0% or 100% pass cannot be flaky, so the two axes share arithmetic. In 72% of null simulations, the correlation is at least as negative as observed.

WHAT SURVIVES

One quarter of repeated cells change answer. Repeat important tasks, monitor mixed outcomes, and treat high flake as an operational warning.

WHAT DOES NOT SURVIVE

The raw -0.95 cannot be presented as an independent discovery that determinism predicts capability. Most of that line is built into the ruler.



07 COST AND VALUE

Price and capability live on different axes

126x

TRIAL-WEIGHTED SPREAD
The report's specified median

161x

TASK-EQUAL SPREAD
Each task receives equal weight

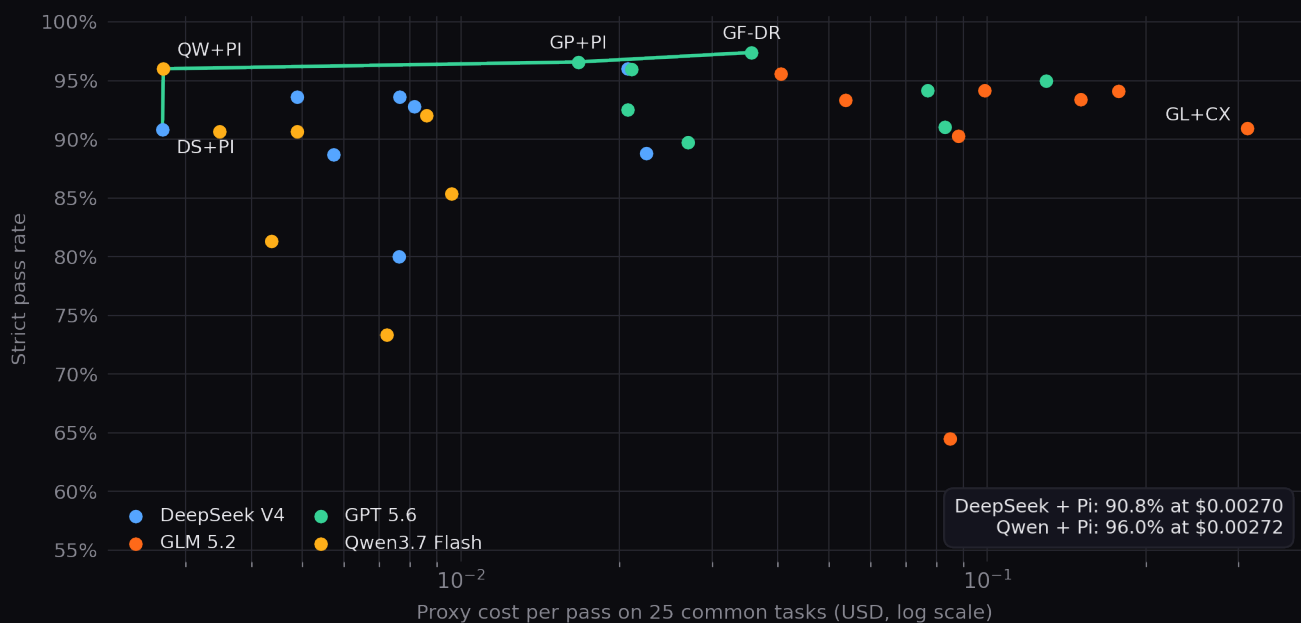
115.5x

COMMON-25 SPREAD
Composition-controlled

$r = 0.219$

COST X PASS
 $p = 0.228$, 32 paid stacks

The value frontier: comparable tasks only



Proxy cost per pass = weighted median task cost / pooled strict rate. Muse is excluded because its contributor edition has no billed cost.

COMPARABLE VALUE FRONTIER

DeepSeek + Pi: 90.8% at \$0.00270 proxy/pass. **Qwen + Pi:** 96.0% at \$0.00272. The first is fractionally cheaper; the second is 5.2 points more accurate. Treat them as a frontier, not a single winner.

DOMINATION

Twelve point-estimate relations are at least 3x cheaper and pass more on the same model, across five base configurations. Bootstrap support ranges widely, so the interface should distinguish **point dominated** from **uncertainty-supported**.



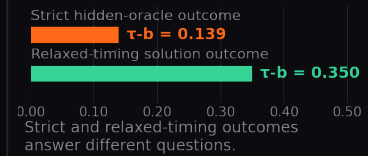
08 HARD WORK

A green suite can hide hard failure

Main-suite rank can move violently on five hard tasks



Easy-to-hard agreement is metric-sensitive



10 / 33

SATURATED TASKS

Pooled strict rate at or above 95%

0.086 to 0.258

BETWEEN-STACK SD

Saturated band vs hard band

THE HONEST INFERENCE

Strict main-suite ranks carry little usable signal about the five hard tasks ($\tau\text{-b} = 0.139$, $p = 0.275$). Relaxed timing raises agreement to 0.350 ($p = 0.0069$). The result is metric-sensitive and the hard band is small.

WHAT SATURATED TASKS ARE STILL GOOD FOR

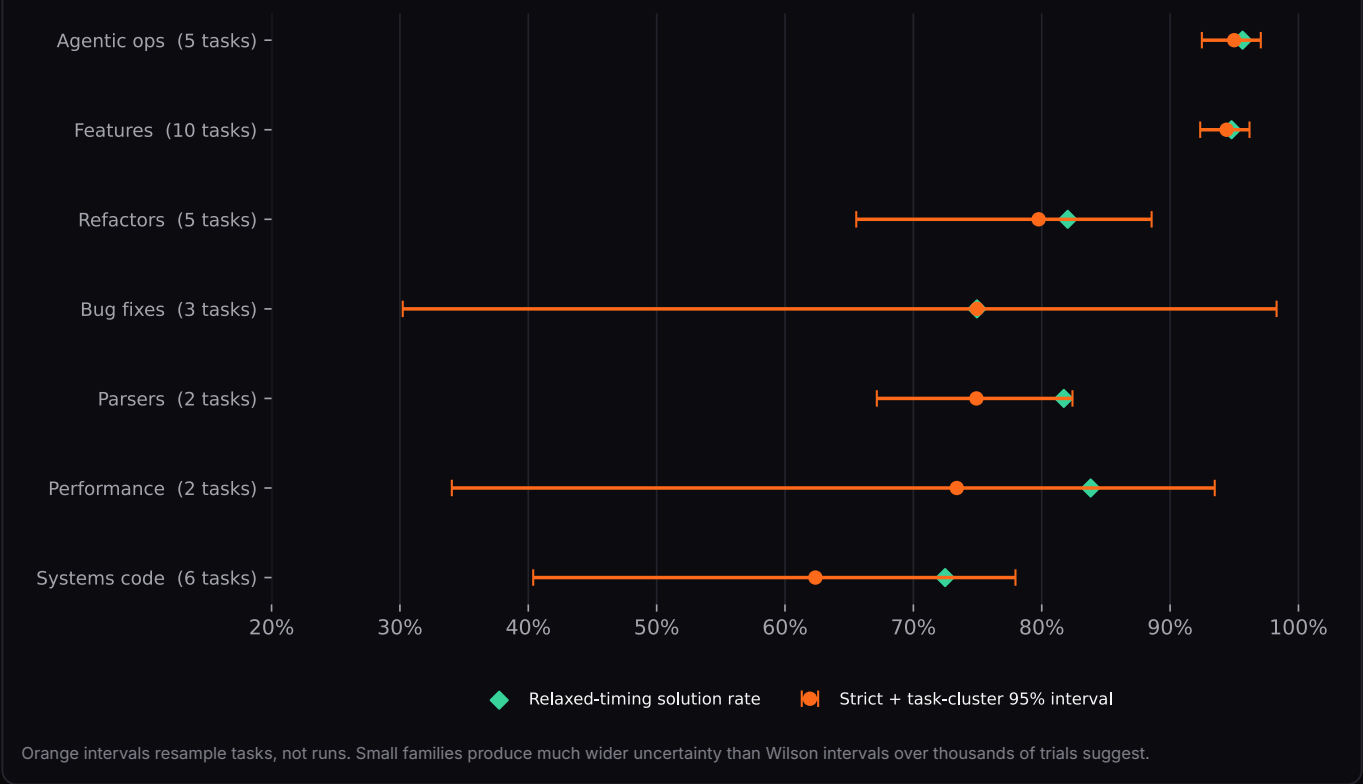
Keep them as regression canaries and minimum-quality gates. Do not let them dominate a leaderboard after nearly every stack passes.



09 CAPABILITY BY WORK TYPE

Strong at the edges, weaker in the core

Capability varies sharply by work family; task-selection uncertainty is wide



FAMILY	TASKS	STRICT	SOLUTION	TASK 95%
Agentic ops	5	95.0%	95.6%	92.5-97.1%
Features	10	94.4%	94.8%	92.3-96.2%
Refactors	5	79.8%	82.0%	65.6-88.6%
Bug fixes	3	75.0%	75.0%	30.2-98.3%
Parsers	2	74.9%	81.7%	67.2-82.4%
Performance	2	73.4%	83.8%	34.0-93.5%
Systems code	6	62.4%	72.5%	40.4-78.0%

MOST RELIABLE

Agentic operations and feature work are about 95% on this task set.

FRONTIER PRESSURE

Systems code is 62.4% strict and 72.5% under relaxed timing. Performance and parser families contain only two tasks each; treat their point rates as indicative.

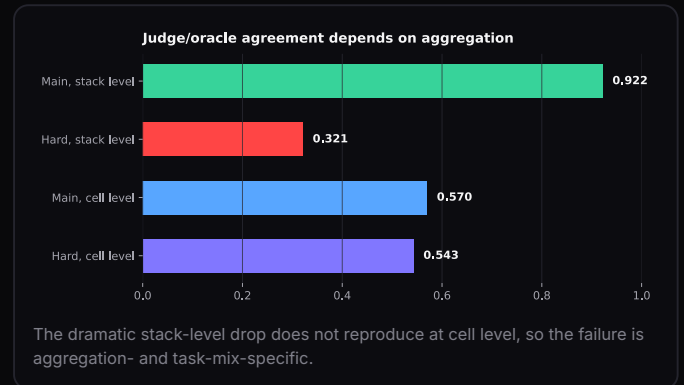
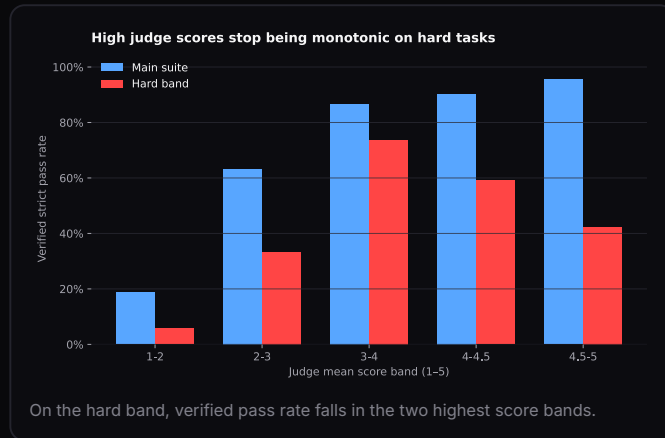
SEPARATE CEILING RESULT

The reported zero-pass 27_ir_optimizer task is absent from the shipped CSV. It should be displayed as a separately sourced ceiling result, not silently included in comparative rates.



10 LLM JUDGE VALIDITY

Good-looking code is not the same as correct code



89.8%

RUN COVERAGE

4,701 of 5,233 judged

0.922

MAIN STACK R

Judge vs strict rate

0.321

HARD STACK R

32 stacks, five tasks

10

DENOMINATOR
MISMATCHES

All on task 03

CONCRETE MISMATCH

GPT 5.6 + Hermes averages **4.05 / 5** for judged quality on the hard band while passing **1 of 15** strict attempts.

CORRECT USE

Use the judge to evaluate craft, readability, strategy, and triage. Use executable oracles to determine correctness. Do not collapse the two into one score.



11 OPERATING PLAYBOOK

How to use BurnBench

- 01 Benchmark the exact stack.** Preserve model, harness, configuration, task family, and outcome definition in every comparison.
- 02 Default to comparable tasks.** Use the common-25 set for cross-model cost and capability; label as-run results composition-sensitive.
- 03 Repeat consequential tasks.** Report the mixed-cell rate beside pass rate. Single-run scores hide a quarter of observed instability.
- 04 Separate cost estimands.** Show median cost per task and proxy cost per pass; do not imply successful-run costs are observed.
- 05 Use two uncertainty layers.** Wilson intervals describe run noise inside a cell. Task-cluster intervals describe sensitivity to which tasks were chosen.
- 06 Keep executable tests in the loop.** Judge scores are useful metadata, not a replacement for correctness oracles on hard work.

BOUNDARIES

Five models; eight harnesses; 33 measured tasks; five hard tasks; Qwen absent from the hard band; list-price-derived costs; no trial-level pass-linked cost, token, latency, or noop records.

INTEGRITY FLAGS

Twenty-three cells covering 73 runs have zero or missing token accounting. Excluding them raises global strict rate from 83.9% to 85.0% and moves some rankings. Ten task-03 cells retain more judge scores than valid trials.

SOURCE OF TRUTH

Primary data: **burnbench-consolidated.csv**

SHA-256:

289d9695757c189d0e2a8038c0755c70b1c4bd208c4aa1c5fb6d548f9b570849

The model matters. The harness matters more. The pairing is the product you actually deploy.